

1 Local interaction networks reconstructed from global biodiversity data improve 2 pollinator restoration decision making

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31 Abstract

32 Global pollinator declines threaten the health of ecosystems and food systems, underscoring
33 the urgency of conservation actions such as habitat restoration. However, data gaps on plant
34 use among pollinators continue to limit reliable design of restoration plant mixes. To address
35 this, we present NECTAR (Network-Enhanced Conservation Tool for Analysis and
36 Recommendation), a new modular framework that integrates multiple data modalities—including
37 species distributions, phenometrics, and phylogenetic data—to infer flower visitation and host
38 plant interactions from spatial, temporal, and phylogenetic overlap, generating spatially explicit
39 plant-insect interaction networks that guide planting recommendations for pollinator habitat
40 restoration. We demonstrate the utility of NECTAR by generating a large plant-insect metaweb
41 across California, comprising 1,247,081 spatially explicit interactions for 5,131 pollinator species
42 and 5,178 native plant species. Predicted networks recovered significantly more interactions
43 than null models, demonstrating that integrating multiple ecological constraints improves
44 interaction prediction. In realistic restoration simulations, NECTAR's data-driven plant mix
45 recommendations support up to 2.8 times more pollinator species compared to existing

resources and random selection of plants. This optimization facilitates the inclusion of multiple goals and constraints, and provides complementary decision-making information to existing resources. NECTAR offers a scalable, evidence-based framework for translating increasingly available global biodiversity data into locally actionable restoration guidance, with broad potential to improve pollinator habitat restoration worldwide.

Main

Global insect declines^{1,2} are compromising ecosystem functions and services that sustain human health, economic security, and ecosystem stability^{3–7}. Foremost among these is pollination, a service performed largely by insects which underpins reproduction in up to 90% of flowering plants species⁸ and is integral to global food security^{9–11}. With more than one in five pollinators at risk of extinction¹², scientists have made urgent calls for policymakers, conservation practitioners, and the public to prioritize pollinator conservation^{13–15}. Since the loss and degradation of native habitat is one of the primary threats to insects^{16,17}, habitat restoration or enhancement (hereafter restoration) has become a central pollinator conservation strategy worldwide—one to which even small-scale actions by individuals can meaningfully contribute. Efforts to increase the quantity and quality of habitat have been shown to provide substantial benefits to pollinator communities^{18–21}, including in anthropogenic landscapes²². With growing public awareness of insect declines and interest in pollinator gardening²³, there is currently an unprecedented opportunity to engage diverse private and public land stewards in an “all hands on deck” approach to counter pollinator declines through habitat enhancement.

The value of habitat restoration for pollinators, however, depends on the strategic selection of plants that meet their specific resource requirements²⁴. Many insects have evolved to be dependent on narrow sets of plants^{25,26}. At a given location, for example, an average of 69% of caterpillar species feed only on a single plant family²⁵ and 50% of bees provision pollen from only one or two plant families, with some restricted to only a single plant species²⁶. While native plants are beneficial to pollinators^{27,28}, individual plant taxa differ widely in the number and identity of insects they support²⁹, so optimal plant mixes should reflect specific restoration goals. For instance, plantings designed to conserve individual threatened or highly specialized pollinator species may prioritize specific host plants, whereas those intended to maximize ecosystem services such as crop pollination may favor plant species that support the greatest number of pollinator species^{30–32, 12}. These goals do not always align, and random selection from the regional plant species pool cannot be assumed to achieve either^{33,34}. Detailed knowledge of which plants support which pollinators would greatly facilitate pollinator habitat optimization, enabling practitioners to tailor plant mixes to specific goals rather than relying on generalized assumptions about nativeness alone.

A growing array of plant lists, guides, and restoration tools have greatly improved support for pollinator-friendly planting by providing generalized recommendations for broad audiences. While these resources often reflect substantial ecological expertise, they are designed for general audiences across diverse landscape contexts, and therefore lack the detail and flexibility required to meet the variety of constraints and goals that practitioners and gardeners

face in individual projects³⁵. Translating these lists into effective site-specific plantings typically requires additional specialist knowledge, limiting the ability of the many gardeners, farmers, and other land stewards without that expertise to maximize the conservation impact of their efforts^{36–38}. For instance, existing resources rarely specify which pollinator species are supported by each plant species or how this varies geographically, limiting their utility for projects with explicit small-scale community- or species-level restoration goals^{35,39}. Due to this lack of comprehensive ecological data, such lists have been shown to provide incomplete and inconsistent planting recommendations that omit many effective species, include suboptimal ones, and favor charismatic, easily identifiable pollinators (e.g., butterflies and bumble bees)^{40,41}. Together, these limitations highlight the need for complementary tools that translate ecological data into context-specific, evidence-based guidance—customizable to the goals (e.g., target species conservation or ecosystem service optimization) and practical constraints (e.g., soil type, irrigation requirements, and nursery availability) that characterize every restoration, landscaping, and conservation project^{39,42}.

Data-driven, network-based approaches offer a promising approach to identifying plant species that support high pollinator diversity and promote community stability^{43–46}, while also providing quantitative criteria to support multi-objective decision making^{33,46–50}. Network-based approaches have been used to optimize pollinator plant mixes at local scales^{34,51}, but remain as conceptual frameworks in broader applications due to a lack of comprehensive data on biotic interactions (the Eltonian Shortfall of biodiversity data)^{52,53}. Implementing network-based approaches at broader scales thus requires deriving reliable interaction networks from large-scale biodiversity data that are increasingly available from community science contributions^{54–58}, archived datasets, and digitized museum specimens^{59–62}. However, these data can be biased toward charismatic and easily identifiable species and toward more accessible locations^{63–68}, such as roads⁶⁹. There is thus an urgent need for robust analytical frameworks that can derive interaction networks from heterogeneous biodiversity data, account for inherent biases, and translate the results into actionable, spatially explicit guidance for pollinator habitat restoration.

In response to this need, we present NECTAR (Network-Enhanced Conservation Tool for Analysis and Recommendation), a new modular ecoinformatic and statistical framework that integrates multiple data modalities—including species distributions, phenometrics, and phylogenetic data—to generate spatially explicit plant-pollinator interaction networks and plant recommendations given planting constraints and objectives. To test whether NECTAR improves the pollinator conservation value of planting recommendations, we applied the framework in California, one of the world’s floral and entomological hotspots, generating one of the largest spatially-explicit plant-insect metawebs to date with over 1.2 million spatially explicit interactions for over 400,000 plant-insect pairs (including both flower-visitation and host plant interactions) across 35 Jepson eFlora districts⁷⁰ (hereafter referred to as ecoregions). Predicted networks recovered significantly more interactions than null models, demonstrating that integrating multiple ecological constraints improves interaction prediction. As we show below, the resulting data-driven plant mix recommendations support higher pollinator richness than existing pollinator plant lists when compared using the predicted networks from NECTAR, suggesting that the predicted networks help mitigate gaps in underlying interaction data. When compared

using a metaweb of previously observed interactions alone, the plant recommendations from NECTAR performed comparably to existing lists, suggesting that existing lists are leveraging known interactions well. NECTAR also facilitates the optimization of additional goals in realistic gardening and restoration scenarios by allowing the integration of other information alongside the quantitative values of pollinator support. To enable public access to the recommendations provided by NECTAR in California, we applied the results in the development of the Calscape Pollinator Companion tool (Box 1), a pollinator plant selection tool for habitat gardening. Because of its modularity, the framework can be extended to other regions, planting contexts, goals, and interaction types, such as at-risk-species conservation, companion planting in agriculture, or supporting natural enemies. We conclude by discussing the minimum requirements needed to extend NECTAR to other regions and possible extensions to support restoration decision making in other contexts.

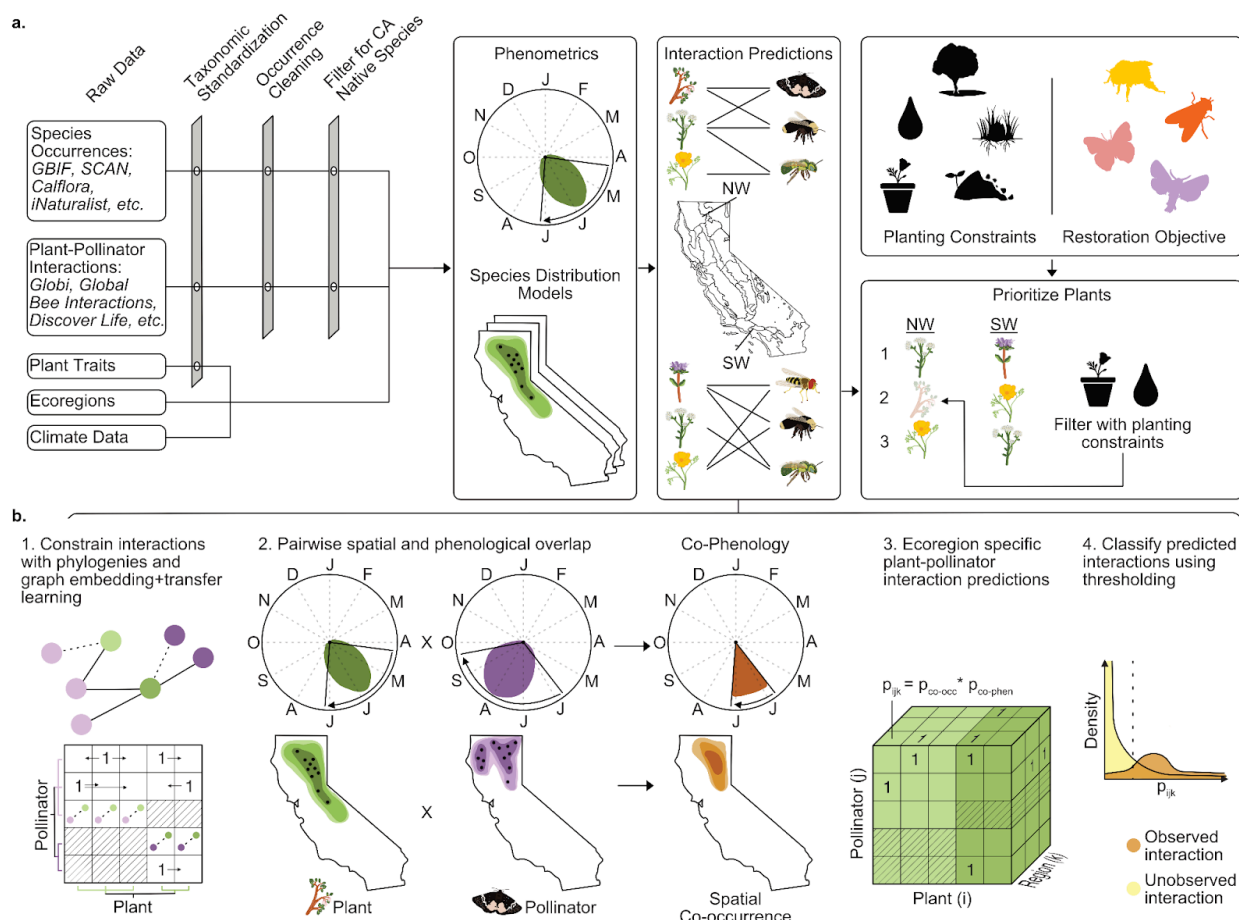


Figure 1. Application of NECTAR to provide regional planting recommendations in California. NECTAR's modular design includes a data aggregation and interoperability module, a network prediction module, and an optimization module. **a.** Conceptual workflow for integrating species occurrence data, plant-pollinator interaction records, and environmental variables to predict spatially-explicit plant-pollinator interaction networks across California's 35 ecoregions (Jepson eFlora districts). **b.** We predicted spatially-explicit networks by combining estimates of spatial co-occurrence (derived from ensemble species distribution models) with phenological overlap between plant flowering and pollinator flight

periods (estimated using circular statistics). We identified potential plant-pollinator interactions using phylogenetic information (genus-level matching for taxa with existing interaction records) and/or graph embedding with transfer learning (for taxa lacking interaction data), with the primary goal of this filtering step to remove biologically implausible pairings based on observed interaction networks (b1). For all potential interaction pairs, we then calculated the spatial overlap (for both flower visitation and Lepidopteran host interactions) and phenological overlap (for flower visitation only) across all ecoregions (b2) and calculated the plausibility of a plant-pollinator interaction occurring in a given region as a function of spatial and phenological overlap (b3). Finally, we classified interactions as highly plausible using a threshold based on the 25th percentile of confirmed spatially-explicit interaction probabilities (b4; see also Supplementary Material 2 Fig. S1). These interaction networks are then integrated with restoration/planting constraints (as we exemplify below) to identify spatially-explicit plant recommendations for pollinator restoration.

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169 Results

170 Overview of NECTAR

NECTAR is a modular framework that integrates large-scale biodiversity data and ecological network modeling to identify spatially-explicit native plant recommendations that optimize ecosystem restoration goals subject to planting constraints (Fig. 1a). NECTAR is available as a pipeline (github.com/EntoDataSciCornell/NECTAR) as well as implemented for the general public in California (Box 1). We applied NECTAR to predict pollinator flower visitation and larval Lepidopteran host-plant use networks across 35 ecoregions in California. Using occurrence and interaction data, we identified likely plant–pollinator interaction pairs by filtering the set of all potential pairs using a phylogenetic constraint based on previously observed interactions, and by predicting new pairs using graph embedding with transfer learning⁷¹. We then computed the expected spatial co-occurrence (for both host-plant and flower visitation interactions) and overlap of key life stages (only for flower visitation interactions using flowering phenology for plants and adult flight periods for pollinators) in each ecoregion to compute an interaction plausibility score and classified likely unobserved interactions using a threshold based on the plausibility scores of observed spatially-explicit interactions (Fig. 1b). Finally, we used optimization approaches including genetic algorithms to maximize pollinator richness and beta diversity with a small set of plant choices.

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188 Constructing spatially-explicit plant-pollinator networks with NECTAR

We aggregated occurrence and interaction data for tens of thousands of species, resulting in 5,391 plant species and 6,242 pollinator species native to California (1,610 bees, 214 hoverflies, 281 butterflies, and 4,137 moths, Table 1.). Among these species, 5,178 plants and 5,131 pollinators had at least one georeferenced and dated record to estimate both species distributions and phenometrics using our tiered approach (and thus included in our interaction prediction framework, see Methods; Supplementary Material 1 Table S1 and Supplementary Material 2 Figs. S2-8). We compiled 287,231 unique interaction records from global biodiversity databases (e.g., Globi⁵⁹) and through data scraping (see Methods). These records included repeated plant–pollinator pairs, but varied in the quality of geographic information, and differed in taxonomic resolution. These 287,231 records consisted of flower visitation by 2,116 pollinator

species on 1,905 plant species (804 genera) and larval host plant use by 242 butterfly and 1,940 moth species on 1,475 plant species (499 genera) native to California (Supplementary Material 2 Fig. S9). In total, there were 51,039 records of unique plant-pollinator interaction pairs, of which 26,007 identified both plant and pollinator at the species level with both species native to California. The remaining interaction pairs were either at a lower taxonomic resolution (e.g., plant identified to genus level) or either the plant or pollinator was not a known California native (Supplementary Material 2 Fig. S10).

Table 1. Summary of the occurrence and interaction data used in NECTAR for our analysis in California and the breakdown of interactions predicted by taxonomic group.

	Total	Bees	Hoverflies	Butterflies	Moths	California Native Plants
Number of Species Included	10,309	1,534	202	256	3,139	5,178
Number of cleaned occurrence records	10,608,318	941,707	73,389	498,324	397,114	4,813,147
Number of cleaned interaction records	287,231	163,648	17,219	34,063	72,301	—
Number of unique cleaned interactions at species level for pollinators in our checklist	26,007	14,592	736	5,816	4863	26,007
Predicted number of unique flower visitation interactions	1,247,081	580,378	74,163	225,868	366,672	1,247,081
Predicted number of unique host plant interactions	189,293	—	—	50,913	138,379	189,293

We used the interaction prediction workflow detailed in the Methods to predict 216,513 possible unique plant-pollinator interactions at the species level for flower visitation and host plant interaction pairs between 1,526 pollinators and 3,995 plants (604 genera). Predicted interaction pairs occurred on average in 4.13 ecoregions among potential interaction pairs identified from existing interaction data alone (using phylogenetic constraints, see Methods). Together with graph embedding and transfer learning (see Methods), we also predicted 116,260 unique flower visitation interactions for 817 species of pollinators that have no existing interaction data (with 3,592 plants belonging to 540 genera). These steps alone increased the number of

species-specific plant-pollinator pairs by a minimum of 12X (from 26,007 unique plant-pollinator pairs at the species level in the raw data, to 332,773 predicted). Of the initial 287,231 interaction records, 153,568 had geographic information, and of those, 5,745 were recorded at the species level for both plant and pollinators, were for species native to California, and had a georeferenced locality in California. Because our interactions are spatially explicit, our prediction space included up to 929 million potential spatially explicit interactions ($5,131 \times 5,178 \times 35 = 929,891,130$). Of these, we predicted 1,247,081 plant-pollinator-ecoregion combinations to occur, thereby increasing the number of species-specific and spatially explicit interactions by over 200X.

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228 Comparison of NECTAR and existing planting recommendations

229 We evaluated whether plants identified by our workflow improve expected pollinator support by
230 comparing our recommendations and existing pollinator restoration resources to randomly
231 selected regionally occurring native plants using an iterative stochastic subsampling process. In
232 short, for every region we (i) drew a random selection of plants native to the region to provide a
233 baseline of the minimum percentage of the pollinator community supported; (ii) drew a random
234 mix of 10 plants from expert-based plant lists^{72,73}, which were chosen because they are the best
235 available resources that most closely match the goals of our framework (i.e., optimizing the
236 pollinator value of plantings); (iii) drew a random mix of 10 plants from the recommended plants
237 by NECTAR that on their own support a high richness of pollinators (here, the length of the
238 NECTAR plant list corresponds to the length of the plant list from the expert-based lists in each
239 region); and (iv) identified an optimal mix of 10 complementary plants (i.e. that together
240 maximize pollinator richness) using a genetic algorithm within NECTAR (see Methods). On
241 average across California, 12.8% (95% CrI [11.4%, 14.8%]) of the pollinator community would
242 be supported when selecting plants at random (Fig. 2a - circle), 12.3% (95% CrI [8.2%, 18.4%])
243 when selecting plants from external resources (Fig. 2a - triangle), 27.2% (95% CrI [24.7%,
244 30.7%]) when selecting from high-degree plants at random from lists generated by NECTAR
245 (Fig. 2a - square), and 34.3% (95% CrI [31.3%, 38.1%]) when selecting plants with the
246 optimized approach generated by NECTAR (Fig. 2a - diamond; also see Model Summaries in
247 Supplementary Material 2 Table S1). Based on raw interaction networks alone, we found that
248 plant recommendations identified with NECTAR and the external expert-based plant lists are
249 both significantly better than random (Supplementary Material 2 Fig. S11, Table S2), suggesting
250 that existing lists effectively leverage available data. We also found that the pollinator support
251 quality of plants identified by NECTAR is robust to regions with low density of interaction records
252 (Fig. 2b,c,d, mean slope of expected pollinator support w.r.t. log of interaction data density -0.14
253 with CrI [-0.39, 0.15] for NECTAR complimentary selection to optimize support for all pollinators
254 and mean slope -0.12 with CrI [-0.37, 0.16] for NECTAR rank draws selection to optimize
255 support for all pollinators), while existing resources are not (mean slope of expected pollinator
256 support w.r.t. log of interaction data density was 0.06 with CrI [0.06, 0.07]). This discrepancy
257 reinforces the need to account for spatial gaps in interaction data. Further, our data-driven
258 approach to selecting plant lists for pollinator restoration with NECTAR is expected to increase
259 support across all pollinator groups included in our analysis (Fig. 2a). Although moths, the most
260 speciose pollinator group in our analysis, had the smallest absolute increase in the expected
261 proportion of the community supported with optimized plant selection compared to random

(+13.7%), they had the greatest relative improvement (+273%) compared to other groups (bees: 161%, butterflies: 166%, hoverflies: 215%).

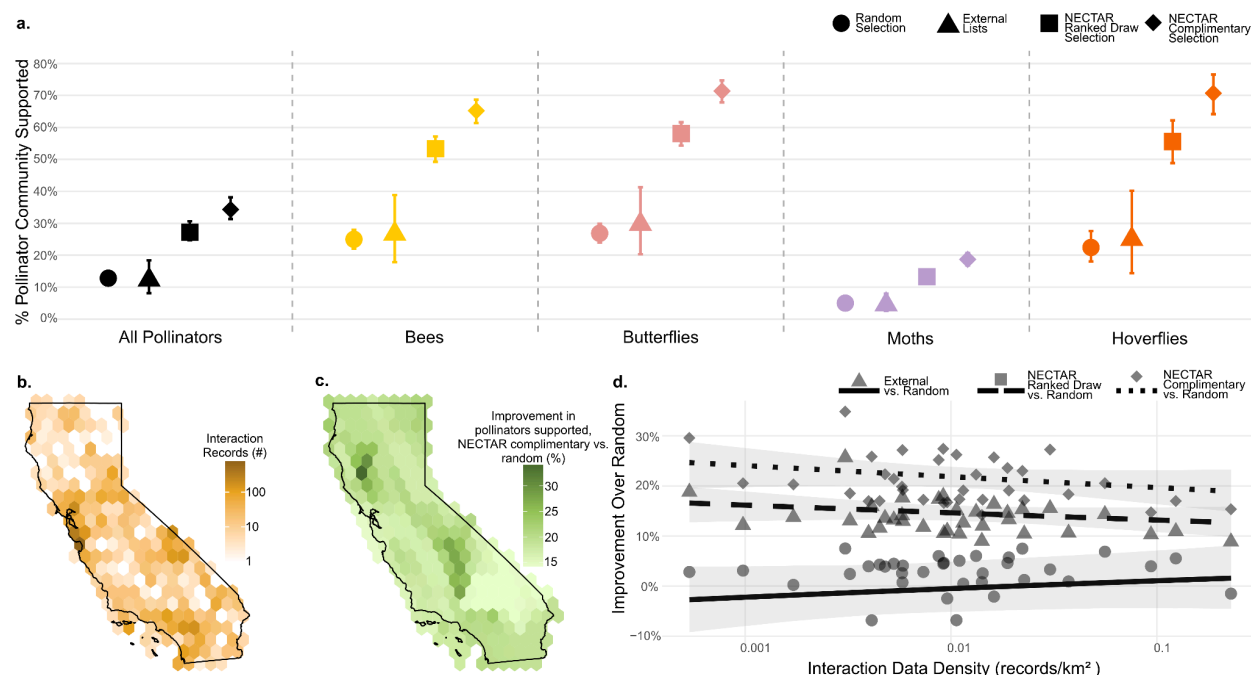
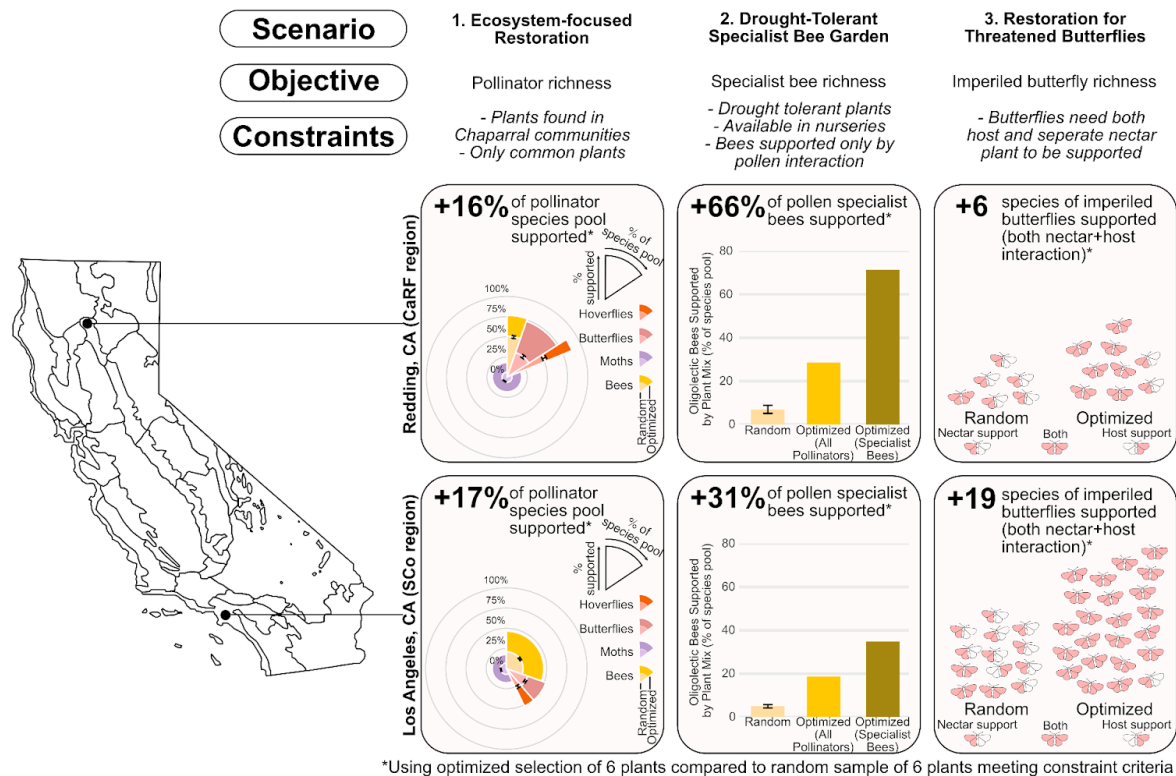


Figure 2. a. The proportion of the pollinator community expected to be supported by 10 plants selected increased as we shifted from: (i) random selection from the pool of regional native plants (circle), (ii) random draws from external expert-opinion based plant lists (triangles), (iii) random draws from the top ranked pollinator support plants we identify with NECTAR (squares), and (iv) the optimal set of complimentary pollinator support plants identified with NECTAR (diamonds). When prioritizing plants for different taxonomic groups we found the same pattern, but with differing levels of support by random plants, where moths are generally the least supported while hoverflies and butterflies are the most supported. Error bars show 95% credible intervals. b. The density map of spatially-explicit plant-pollinator interaction data across California was higher around urban centers such as San Francisco and Los Angeles (also see Supplementary Material 2 Fig. S12). c. The expected absolute increase in total pollinator richness supported by an optimized set of 10 plants compared to a random selection of plants across California was greater around areas with less interaction data. d. Trends of proportion of pollinator community supported across the gradient of interaction data density, showing that NECTAR (squares from the random draws of top degree plants and diamonds from the optimized selection) consistently supports a higher proportion of the pollinator community compared to random, while the improvement of the proportion of the pollinator community supported by external expert-opinion compared to random hovers around 0%. Points correspond to a comparison within one of the 35 ecoregions in our analysis, and lines show the marginal mean relationship between interaction density and increase in pollinator community support. Colors in this figure mirror color for taxonomic groups in all figures below. Ribbons represent 95% credible intervals.

Pollinator restoration under planting constraints

We defined three hypothetical restoration scenarios with varying constraints and objectives, and used a genetic algorithm (see Methods) to identify the optimal set of 10 complementary plants

(those that together maximize support for the local pollinator community) and compared expected pollinator support against random plant selection across California. However, for demonstration purposes, we visualize here for two locations: Redding (Cascade Range Foothills, top) versus Los Angeles (South Coast, bottom). For the rest of California, see Supplementary Material 3 Tables S1-S2 for the optimized plant lists and potential pollinator support. Across all scenarios, data-driven plant mixtures substantially outperformed randomly assembled mixes in terms of the diversity of pollinators supported, while still satisfying the other criteria (Fig. 3). In Scenario 1, our objective was to identify common plants (by removing rare plants, as defined by the California Natural Diversity Database (CNDDB)⁷⁴) that naturally occur in Chaparral habitats which support high richness of all pollinator groups. In general, a greater proportion of bees, butterflies, and hoverflies were supported by optimized plant mixes compared to random selection of plants, with all taxonomic groups seeing significant relative improvement in support for the optimized mix (123% for bees, 142% for butterflies, 176% for moths, and 139% for hoverflies). In Scenario 2, our objective was to identify a garden plant mix composed of commercially available native plants (as defined by the California Native Plant Society) that require little water that best support pollen-specialist bees. We found that a random plant mix generally provided poor support for oligolectic bees, whereas an optimized list identified the regionally important plants for these species, sometimes supporting more than double the number of local oligoleges even compared to the plant list optimized for all pollinators. For Scenario 3, our objective was to identify plants that offer both host and nectar support for declining Southwestern butterfly species⁷⁵. On average across the state, we found that an optimized mix offers a ten-fold increase in the number of species supported with both a host plant and a separate nectar plant compared to random (Fig. 3, Supplementary Material 3 Table S2), demonstrating that a data-driven approach can offer substantial benefit to targeted restoration efforts of imperiled species.



316

317 **Figure 3.** Across three restoration scenarios, we demonstrated that optimization based on pollinator
318 support data can outperform random plant selection by 2-10X, even when both approaches use the same
319 planting constraints (e.g., drought-tolerant, nursery-available plants). For each scenario, we quantified the
320 improvement of data-driven optimization over random selection from the regional constrained plant pool.
321 In (1) *Chaparral restoration scenarios* for two regions (Los Angeles/Redding), random selection left gaps
322 in pollinator support, while optimized mixes supported a higher proportion of all taxa. Insets show grouped
323 radial bar charts where the angular width of each slice (i.e., the width of the pie slice) represents the
324 proportion of that pollinator group in the regional pollinator pool, and the radial extent (i.e., how far the pie
325 slice extends outward from the origin) reflects the proportion of that taxa supported. The lighter inner slice
326 represents pollinator support by random plant selection and the darker outer slice represents optimized
327 selection, with slices independent (not summed) in radial extent. (2) *Specialist bee conservation*: General
328 pollinator optimization misses key specialist supporting plants; targeted optimization for oligolectic bees
329 nearly doubles specialist support on average. The percentage of known specialist bees supported
330 through presumed pollen interactions by random selection, optimization for all pollinators, and
331 optimization to support pollen interactions of oligolectic bees is shown in the inset. (3) *Declining butterfly*
332 *recovery*: Random mixes can provide nectar support but miss host plants; optimization delivers 10X more
333 host plant coverage while also maintaining or increasing nectar availability. The list of identified
334 complementary plants for each scenario above and an unconstrained scenario can be found in
335 Supplementary Material 3 Table S1, and the corresponding expected pollinator support for these plant
336 mixes can be found in Supplementary Material 3 Table S2.

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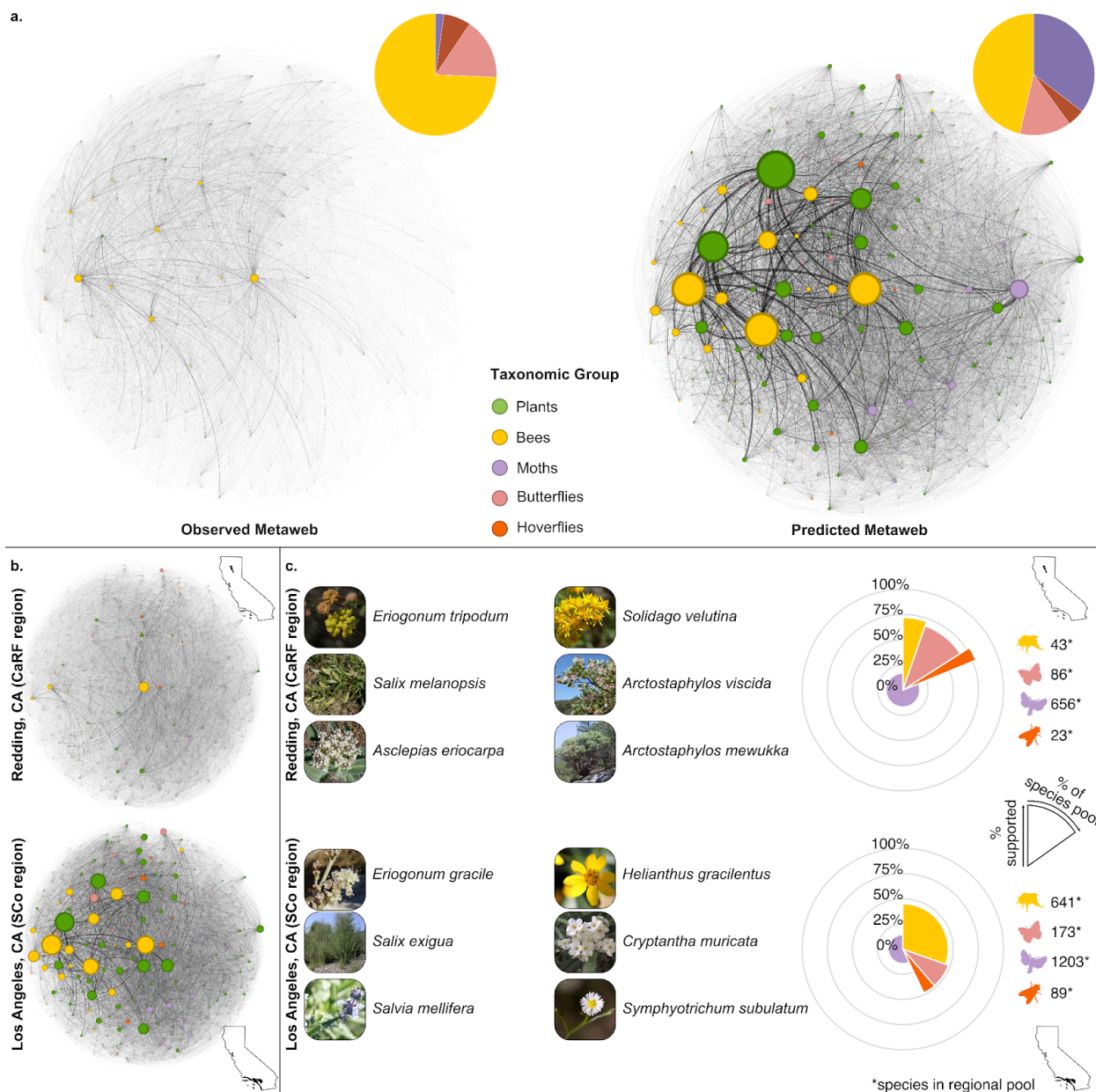
338 How leveraging interactions enables effective restoration decision making

339 Our interaction prediction framework performed well under cross-validation, recovering 53.8% of
340 withheld local interactions on average; significantly exceeding both a fully random null model

341 (0.1%) and a phylogenetically constrained null model that assigns interactions at random among
 342 co-occurring species within known host genera (25.2%; Supplementary Material 2 Figs. S1,
 343 S13-14). The higher recovery rate by incorporating prior information from observed interactions
 344 supports NECTAR's ability to substantially improve inferred network structure. Whereas
 345 observed interaction datasets were heavily biased toward a small number of well-sampled bee
 346 genera, predicted flower visitation networks substantially expanded interaction representation
 347 for moths (i.e., the most species-rich pollinator group in our analysis); thereby capturing a
 348 broader spectrum of pollinator life histories and resource dependencies (Fig. 4a).

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350 Downscaled regional interaction networks also exhibited pronounced spatial heterogeneity (Fig.
 351 4b), driven by high turnover in plant and pollinator species composition, interaction identities,
 352 and overall network structure across ecoregions (Supplementary Material 2 Figs. S15-S22).
 353 This spatial variation fundamentally altered which plant species contributed most to supporting
 354 pollinator communities and how many pollinator species each plant supports. As a result, plant
 355 species that are highly valuable for restoration in one region can provide limited benefits in
 356 others. This provides evidence that network structure may further mediate restoration efficiency:
 357 in more nested networks, a smaller subset of plant species supports a larger proportion of the
 358 pollinator community. Together, these results show that explicitly modeling plant–pollinator
 359 interactions across space reveals spatially contingent restoration priorities that cannot be
 360 captured by plant occurrence data alone.



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Figure 4. Predicted interactions expand taxonomic coverage and reveal spatial variation in network structure that helps inform location-specific plant recommendations. **a.** Observed versus predicted flower visitation metawebs demonstrate improved representation across pollinator taxa. Our predictions (right) greatly increase representation of moths (purple), the most diverse but under-sampled pollinator group, while maintaining the core interaction structure present in raw observations (left). Node size represents genus-level degree (number of species interaction partners within that genus); link width represents the number of unique species-level interactions between genera. Both networks are scaled identically for comparison. **b.** Local networks differ dramatically in structure across California. Genus-level flower visitation networks for Redding (Cascade Range Foothills, top) versus Los Angeles (South Coast, bottom) show substantial differences in network size, composition, and connectivity patterns. Redding's network has 26% the number of interactions as the network around Los Angeles, $\sim\frac{1}{2}$ the number of plant species and $\sim\frac{1}{3}$ the number of pollinator species, and is less nested with a lower average degree (Supplementary Material 2 Figs. S17-18). Furthermore, moths make up a much larger percentage of the pollinator

community in Redding (58% versus 39% in Los Angeles). These structural differences drive spatial variation in the identity of plants that provide the most conservation value to pollinator communities. c. Optimal plant mixes differ by location due to network variation. The six most complementary plants for supporting local pollinator flower visitation interactions in Redding versus Los Angeles do not share any species in common. Radial pie charts show pollinator community composition (angular; i.e., how wide the pie slices are) and the proportion of each group supported by the optimized plant mix (radial; i.e. how far the pie slice extends outward from the origin). These six plants support just over ¼ of the pollinator community in both Redding and Los Angeles, but the relative makeup of who is supported differs substantially, reflecting species pools and network structure differences shown in panel b. Genus level networks visualized with Gephi⁷⁶. Photographs in panel c from [Calscape.org](https://www.calscape.org); *Eriogonum tripodum* by Barry Breckling, *Salix melanopsis* and *Salix exigua* by Keir Morse, *Asclepias eriocarpa* and *Eriogonum gracile* by Neal Kramer, *Solidago velutina* by Stan Shebs, *Arctostaphylos viscida* by Scot Loring, *Arctostaphylos mewukka* by Dean Wm. Taylor, *Salvia mellifera* and *Helianthus gracilentus* by Calscape, *Cryptantha muricata* by National Park Service, and *Symphyotrichum subulatum* by Richard Spellenberg.

Discussion

Building on recent conceptual roadmaps^{49,50} and prior applications of plant–pollinator networks at local scales^{41,77}, this study provides the first working example of how to scale a data-driven, network-based approach to pollinator habitat restoration across large geographic regions. Our novel framework, NECTAR, integrates multiple data modalities to generate spatially explicit plant–insect interaction networks that quantify plants’ ecological value to pollinator communities, which can be used to optimize plant mixes for pollinator conservation alongside other constraints. Implementing this framework using statewide data from California enabled us to generate one of the most comprehensive plant–pollinator metawebs to date⁷⁸. NECTAR’s data-driven plant mix recommendations identified regionally important, high-value plant species overlooked by existing pollinator plant lists, increasing the richness of pollinators supported while enabling simultaneous optimization of multiple, real-world restoration objectives, including drought tolerant habitat design and targeted species conservation. NECTAR showed the greatest improvement over existing lists and random plant selections in data-poor regions and for undersampled pollinator groups, reflecting its ability to correct spatial and taxonomic sampling biases by generating informed interaction predictions from integrated data. Together, these results demonstrate that by explicitly addressing known data gaps and biases and providing quantitative criteria for decision making, NECTAR offers a highly robust and transferable framework for pollinator plant selection across regions and geographic scales.

Our reconstructed networks showed substantially higher recall of withheld interactions than two null models: one assigning interactions at random among all species and another restricting random interactions to co-occurring species within known host genera. These results show that incorporating ecological constraints improves interaction predictions and supports the use of our approach for restoration and conservation. Nevertheless, predicted interactions represent informed estimates rather than direct observations, and may still be skewed by biases in the underlying data. Community science data is heavily biased towards easily-identifiable taxa like bumble bees or butterflies with striking patterns⁶³, and while our results show that data integration and prediction helped address these taxonomic biases, there is still a need for data

collection efforts focused on underrepresented pollinator taxa such as moths. Furthermore, reconstructing spatially explicit subnetworks from a metaweb assumes that species capable of interacting in one context may also interact in other regions where they co-occur spatially and temporally. Here, we downsampled state-level metawebs to regional metawebs⁷⁹, but we did not attempt to downscale metawebs down to realized local scale networks. Realized networks in individual locations depend on both local scale environmental filtering, such as filtering from urban environments⁸⁰, as well as the local biotic context, such as species local relative abundances. For example, insects may exhibit greater specialization at local scales despite being more generalized at broader scales because of relative abundances within plant communities^{81,82}. Future work should prioritize empirical validation of predicted interactions, particularly for those species with little interaction data, including via experimental field manipulations to assess the efficacy of our plant selection optimization criteria⁸³. Despite these limitations, recent global efforts^{84,85} to compile and predict interactions using machine learning can facilitate scaling of this framework, offering a timely approach for enhancing pollinator conservation through revegetation.

The primary goal of this research is to enhance the value of habitat restoration to pollinator communities using data-driven insights. To this end, NECTAR is well-positioned to advance and complement the foundational resources provided by expert-curated pollinator plant lists. These resources^{72,73} are widely adopted by gardeners and restoration practitioners and are often the starting point for public participation in native plant gardening. They reflect practical constraints such as ease of establishment and local nursery availability in addition to plants' pollinator value, and often reflect extensive trial and error along with expert review. When evaluated against a raw metaweb that assumes any previously observed interaction pair can interact wherever both species co-occur, NECTAR's optimized plant mixes were comparable to existing lists in improving pollinator support over random selection (Supplementary Material 2 Fig. S11). This confirms that these resources are, in practice, well-optimized for pollinator support given the available interaction data. However, when evaluated against NECTAR's predicted networks, which address spatial and taxonomic biases in the underlying data, NECTAR's recommendations substantially outperformed existing lists, particularly in data-poor regions and for undersampled pollinator groups. This gap reflects limitations in the underlying plant-pollinator interaction knowledge rather than the quality of expert curation. NECTAR addresses these limitations by filling spatial and taxonomic gaps through species distribution models and interaction prediction, respectively. For example, NECTAR increased the number of flower-visitation interactions involving moths by several orders of magnitude (from only 20 spatially-explicit flower visitation interactions in California to over 360,000 predicted), consistent with evidence that moths interact with a wide diversity of plant species, despite being substantially underrepresented in empirical datasets relative to diurnal pollinators⁸⁶⁻⁸⁸.

NECTAR also provides a quantitative measure of pollinator support, currently lacking in most plant selection resources. This facilitates balancing pollinator conservation with other objectives and constraints often considered in restoration and planting efforts^{34,51,89,90}, including biodiversity goals (e.g., conserving at risk or endangered pollinators, supporting specialists), other environmental and sustainability goals (e.g., reducing stormwater runoff³⁹ or improving climate

resilience), social-cultural preferences (e.g., aesthetics), and practical constraints (e.g., cost, ease of establishment, space, and nursery availability). The ability to weigh pollinator support alongside these other constraints may be particularly useful for maximizing the conservation impact of wildlife gardening and habitat restoration in small spaces, where usually only a subset of the total plant species in a given ecosystem are used⁹¹. To illustrate this, our restoration scenarios explored the optimization of a set of just six plants for projects facing real-world constraints and objectives. For example, recent reports of widespread butterfly declines in the United States⁷⁵ underscore the urgency of butterfly conservation, which requires the coordinated provision of both larval host plants and adult nectar resources to support complete life cycles^{92–94}. By optimizing plant lists using our predicted interaction networks to balance the multiple resource requirements of regionally threatened butterfly species, we increased the average number of threatened species supported with just six plants by an order of magnitude compared to random plant selection.

NECTAR provides a scalable methodological approach with broad potential for global conservation applications. The workflow is modular: researchers can substitute regional occurrence databases, local interaction records, and region-specific spatial units while maintaining the core analytical pipeline. We provide complete code at github.com/EntoDataSciCornell/NECTAR that users can adjust based on local data availability (e.g., stricter interaction probability thresholds for data-rich regions, coarser spatial resolution for data-poor areas). Based on our implementation in California, we found that many species had very few occurrence or interaction records. Generally, more data per species will lead to better spatial and temporal predictions; however, our use of a tiered approach enables the integration of data sparse species. For example, regions with sparser data can still apply the approach by leveraging information about species from the pool from other regions or by aggregating to coarser spatial scales. Recent compilations of global plant-insect interactions^{78,95} and global-scale occurrence datasets⁹⁶ demonstrate that sufficient raw data exist for many regions. For practitioners working in California, immediate access is available through the Pollinator Companion tool (Box 1) and bulk data downloads (doi.org/10.5281/zenodo.19338963).

Beyond the examples explored here, our approach offers opportunities to advance ecological understanding and applied conservation, including: (i) advancing understanding of how ecological interaction networks and species roles vary across space and between different interaction types, which is essential for predicting and mitigating the effects of climate and land use change on biodiversity and ecosystem services^{97–99}; (ii) advancing understanding of drivers of network assembly; (iii) identifying candidate plants with high pollinator value that are not currently available in nurseries, which can be included in propagation trials^{100,101}; (iv) revealing where and for which taxa targeted data collection is still most needed, including identifying gaps where relatively small additions of interaction data could substantially improve network predictions; (v) modelling plant-pollinator networks to identify priority plants under future climate change scenarios, which can be used to advance climate-resilient restoration by considering both future abiotic and biotic conditions; (vi) evaluating pollinator support in an existing communities, and modelling potential change under different vegetation-change scenarios; and (vii) supporting agricultural applications, including identifying companion plants for crop

pollination or natural enemy support, especially given NECTAR's modularity, which allows different interaction types and objectives to be incorporated.

Applications and Implementation

Our results show that our data-driven approach can inform planting decisions to achieve specific pollinator conservation goals alongside other constraints in restoration, gardening, agriculture, landscaping, and related applications. Conservation goals can vary widely from maximizing pollinator richness, to maximizing phenological overlap of nutritional resources⁵¹. Interaction data can be integrated with other planting goals and constraints into a new generation of interactive plant selection tools designed to make evidence-based restoration more accessible to diverse audiences. We present the Pollinator Companion as one example of a data-driven tool for gardeners, farmers, and conservation practitioners that leverages NECTAR to improve the ecological relevance of plant selections (Box 1). NECTAR can be leveraged in other applied projects to build tools tailored to different goals and audiences including farmers, restoration practitioners, and endangered species conservation.

Box 1. Improving plant selection tools with data-driven insights: Calscape’s Pollinator Companion

Translating complex ecological data into conservation action requires tools that can be adopted by diverse end-users with varying priorities, constraints, and levels of experience^{102, 103}. One impactful group of stakeholders are land stewards in urban and agricultural areas, who help support pollinators through gardening and landscaping with native plants. To provide data-driven guidance to these audiences, we partnered with the California Native Plant Society to integrate our framework into Calscape’s Pollinator Companion, a pollinator gardening tool: pollinators.calscape.org/pollinator-companion/welcome. Developed through iterative user testing for accessibility to non-specialist audiences, this interactive tool brings users through a series of steps enabling them to optimize pollinator support with other planting needs or constraints (e.g., drought tolerant plants).



Calscape
California Native Plant Society

1

Region

2

Pollinators

3

Goals

4

Soil Type

5

Sun Exposure

6

Irrigation

Save to a plant list

Print your list

Share your list

Your pollinator planting guide is ready!

Your tailored list is based on your ecoregion, site conditions, and goals. Not all plants recommended for your region may be appropriate for your exact location.

Please select from the plants below to further customize your list.

South Coast

All Pollinators

Full Sun

Your selections

10 of 833

native plants

335

Butterflies and moths

1

Hummingbirds

314

Bees

21

Hoverflies

Supporting 671 of 1666 available pollinators

Selected 10 of 833

Select all | Unselect all

Group by None

Sort by Pollinators Supported

☒



California Buckwheat
Eriogonum fasciculatum

☒



Black Sage
Salvia mellifera

☐



Purple Sage
Salvia leucophylla

☐



White Sage
Salvia apiana

☒



Chia
Salvia columbariae

☒



Wild Heliotrope
Phacelia distans

☒



Coast Live Oak
Quercus agrifolia

☐



Parry's Phacelia
Phacelia parryi

☐



Ashyleaf Buckwheat
Eriogonum cinereum

☒



Hummingbird Sage
Salvia spathacea

17

Figure 5. The California Native Plant Society Calscape Pollinator Companion, available through Calscape.org, presents multiple goals that users can select, including those related to pollinators and other planting priorities (e.g. improving soil health or finding crop pairings), as well as planting constraints due to soil type, irrigation amount, and sun exposure. Ultimately, the Pollinator Companion provides a list of plants that meet the constraints and maximize the user-specified goals. In the display, plants are ordered based on their value to the pollinator community, users can select plants to view lists of supported pollinators, and adding plants interactively changes the pollinator community potentially supported.

Methods

Here, we describe a reproducible and scalable data-driven framework with R 4.3.3¹⁰⁴ and Julia¹⁰⁵ that leverages large-scale biodiversity data to generate spatially explicit, data-driven native plant restoration recommendations for supporting pollinators (NECTAR), and demonstrate the added value of this workflow in California. We did this by first generating a metaweb of all potential plant-pollinator interactions among native plants, bees, moths, butterflies, and hoverflies using observed interaction data for both flower visitation and Lepidopteran larval host plant use. We then downscale to regional interaction networks for California's 35 Jepson eFlora districts using a combination of spatial and phenological overlap between all potentially interacting plant-pollinator pairs (for flower visitation) and spatial co-occurrence alone (for Lepidopteran host plants) (Fig. 1). While we used the Jepson eFlora districts for our application in California, NECTAR can be used for any other spatial criteria (e.g. EPA ecoregions).

We used a multi-tiered approach for estimating both species distributions and phenological distributions of key life stage events (described below) because many species were rare and/or geographically restricted, and thus not able to be modelled with more data-intensive methods. This allowed us to include even species with a single record. While intensive modeling is not appropriate for these species, simpler approaches have been previously used in the literature for insects¹⁰⁶. These rare and geographically restricted species are important to include because they may be targets of conservation, and have important ecological interactions due to the specialized niches they inhabit. Had we not included a convex hull method for estimating the distributions of rare species, or the interval/buffer approaches for estimating phenological distributions, 1,997 pollinators (572 bees, 29 butterflies, 113 hoverflies, and 1,283 moths) and 929 native plants would have been omitted from our analysis, thus removing 72,805 flower visitation interactions between 35,228 interaction pairs and 19,645 larval Lepidopteran host use interactions between 7,920 interaction pairs.

We then used these predicted regional interaction networks to identify the top pollinator-supporting plants and compared them to existing resources to demonstrate the potential improvement for pollinator restoration. Further, we designed several restoration scenarios with various goals and realistic constraints to highlight how a data-driven approach enables rapid integration of constraints and competing restoration objectives. Below is a concise description of our interaction prediction workflow, followed by our methodology for comparing

the top plants identified using our predictive workflow to other restoration resources, designing the restoration scenarios, and estimating added value in each case.

Occurrence data acquisition and cleaning

We used plant and pollinator occurrence records for all native plants, bees, moths, butterflies, and hoverflies in California. Occurrence records for plants were obtained from Calflora¹⁰⁷, a curated database of California wild plants, and the Consortium of California Herbaria¹⁰⁸, an online repository hosting many digitized herbarium specimens across California. Occurrence records for pollinators were obtained from the Global Biodiversity Information Facility¹⁰⁹, Ecdysis¹¹⁰, a curated dataset of bee occurrences for the United States⁹⁶, and Moth Photographers Group¹¹¹. We harmonized and cleaned all occurrence records using established cleaning workflows —namely Biodiversity Data Cleaning (bdc)¹¹² and Bee Biodiversity Data Cleaning (BeeBDC)⁹⁶. Our occurrence data cleaning consisted of harmonizing variable names across datasets, spatial cleaning and filtering for geolocated occurrences within 100 km of California using the tigris¹¹³ TIGER/line delineation of the state, temporal cleaning to extract collection date information from multiple columns and removal of records prior to 1900 (except for Moth Photographers Group data, since date information is not readily available for most records), taxonomic name cleaning with Global Names Parser (GNparser)¹¹⁴, taxonomic harmonization using Global Names Verifier (GNverifier)¹¹⁵ with manual corrections, and record de-duplication.

Taxonomic name harmonization

Any time that we utilized a new dataset in our workflow, we harmonized all taxonomic names. We did this by separating the list of names for plants and animals, since binomial names can be duplicated across Kingdoms, and for each, parsed the list of names with Global Names Parser (GNparser)¹¹⁴. We passed all unique parsed names to Global Names Verifier (GNverifier)¹¹⁵, returning all matches across taxonomic authorities, filtered to only matches that provide higher taxonomic information and then to those that place the species into the correct Kingdom, and from the remaining matches chose the one with the best sort score from GNverifier, which is based on how closely the name matches and the quality of the taxonomic source (manually curated authorities are valued higher than automatically curated and uncurated sources). We then cleaned the matched names with GNparser, and applied manual name corrections (e.g., removed matched species that are known non-natives or merging synonymous species not correctly identified by the matching names produced by GNverifier). Lastly, we harmonized all plant names to currently accepted names according to the Jepson eFlora⁷⁰ using their list of accepted and synonymous plant names, first by trying to match with the original parsed names, and then by the selected harmonized name from Global Names Verifier. We used Jepson eFlora since it is widely considered the most curated and accurate authority for plants found in California.

Species checklists

We generated species-level checklists of native plants and pollinators in each region they occur (Supplementary Material 1 Table S1) to be included in downstream modelling applications by

cross-referencing our cleaned occurrence records with external lists of known non-native taxa^{116–118} and appended additional information on the rarity and endangered status of each species⁷⁴. In all instances of merging information from external lists with our species checklists, we first harmonized all taxonomic names as described above. We filtered out known non-native species for both plants and pollinators from our checklists and occurrence dataset using ref.^{116–118}. For example, we omitted honeybees (*Apis mellifera*) from the checklist of pollinators.

Species distribution modelling

We generated species distribution models for all native plant and insect species using an ensemble-modeling workflow that combined cleaned occurrence records with a set of 50 environmental predictors spanning climate, topography, soils, vegetation, hydrology, and human disturbance, at 5 arcminute resolution. We assembled predictors from WorldClim v2.1 bioclimatic variables, SRTM-derived terrain metrics, SoilGrids soil properties, ESA WorldCover land cover fractions, TerraClimate water balance variables (climatic water deficit, actual evapotranspiration, vapor pressure deficit, solar radiation), MODIS-derived NDVI and leaf area index, GPW population density, the 2018 Human Footprint dataset, and distance-to-coast and distance-to-stream metrics. All layers were downloaded, processed, and aligned to a common 5 arcminute EPSG:4326 grid, with remote sensing products accessed via Google Earth Engine.

For each species, we first spatially thinned occurrences to one record per predictor raster cell, and removed environmental outliers for species with more than 25 occurrences. We then defined an ecologically realistic accessible area by identifying the regions containing the species' occurrences and applying a rarity-scaled buffer. Within this accessible area, we extracted environmental conditions and reduced the predictor set using a two-step procedure implemented via the covsel¹¹⁹ package: (1) we removed highly correlated variables by iteratively retaining the most informative member of each correlated pair, and (2) we ranked the remaining variables using embedded selection methods within generalized linear models, generalized additive models, and regularized random forests to identify the top predictors for final modeling. We generated background points using a target-group sampling strategy, drawing from occurrence records of other species within the same taxon that fell within the species' accessible area, thinned to one point per raster cell. A total of 5,000 background points were sampled for rare species (ESM models) and 10,000 for common species (ensemble models). We then fit models appropriate to each sample-size class: alpha-hull or buffered range models for ultra-rare taxa ($1 \leq n < 7$), ensemble of small models (ESM) for rare taxa ($7 \leq n < 25$), and full ensemble models (GLM, GAM, Random Forest, and MAXNET) with spatial block cross-validation for common taxa ($n \geq 25$). For taxa with ≥ 7 occurrences, we constrained extrapolation using multivariate environmental similarity (MESS) analyses and applied dispersal-limitation post-filters when occurrences were strongly clustered. Finally, we masked all continuous predictions to the Jepson eFlora districts and converted them to binary range maps using the 10% training omission threshold (OR10). Outputs from our species distribution models are available at doi.org/10.5281/zenodo.19338963.

Phenometrics

Because many California species are active during winter when rainfall peaks, we treated the year as cyclical to avoid inflated estimates when active periods overlap the turn of the year, a common limitation of linear day-of-year methods (e.g., phenesse¹²⁰). To estimate flowering periods for angiosperms and flight periods for adult insect pollinators, we used a three-tiered approach based on data availability at the relevant life stage. For species with >10 unique records on ≥ 3 unique days of the year (tier 1), we fit sine-skewed von Mises distributions¹²¹ to circular-transformed observation dates using maximum likelihood estimation and defined onset and offset as the 5th and 95th CDF percentiles (~90% of observations). For species with <10 records on ≥ 2 unique days (tier 2), we used the minimum circular interval that encompasses all observations. For species with records on only one unique day (tier 3), we applied a ± 7 -day buffer. All species had their phenometrics estimated with only one of the tiered methods. This buffer was chosen to retain very data-limited species in the analysis while remaining conservative. Since narrower phenological overlap reduces predicted interactions in NECTAR, underestimates of phenological breadth for a given species will tend to underestimate predicted interactions. Because species could either be bivoltine or flower multiple times, we ignored the peak of the distribution and only estimated onset and offset; for those species, we are potentially overestimating duration and thus interactions.

Tier 1 estimates were calculated at four nested spatial scales (Jepson districts, regions, provinces, and statewide; requiring ≥ 10 observations on ≥ 3 unique days), with each district assigned the finest-resolution estimate available using a hierarchical approach, defaulting to successively coarser scales when district-level data were insufficient. However, for regions with sparse data, estimates were derived from aggregated data spanning multiple Jepson eFlora districts, which likely inflated estimated phenological breadth. Tiers 2 and 3 estimates were calculated statewide and applied uniformly across a species' known range; note that tier 1 estimates for data-poor regions may also overestimate phenological breadth due to aggregation across regions. For insects, only adult records were retained; for plants, we used annotated flowering records from the Consortium of California Herbaria and Phenovision¹²², a machine learning method for identifying flowering records from iNaturalist images. Phenometrics for all species-regions included in our analysis are available at doi.org/10.5281/zenodo.19338963.

Interaction data acquisition and cleaning

We compiled flower visitation records for all pollinator taxa and host plant records for Lepidoptera from multiple sources, including the open-access databases Global Biotic Interactions (GloBI)⁵⁹, the Natural History Museum's HOSTS Lepidoptera hostplant database¹²³, and CropPol¹²⁴. Using custom R scripts, we also scraped interaction records from the Discover Life website¹²⁵, Catalog of Hymenoptera in America North of Mexico, Volume 2¹²⁶, California Plants as Resources for Lepidoptera: a guide for gardeners, restorationists and naturalists¹²⁷, and Pollen Specialist Bees of the Western United States¹²⁸. Finally, we integrated butterfly host plant records supplied by Doneski et al.¹²⁹, a dataset compiled by integrating dozens of review papers, more than 100 field guides, online databases such as HOSTS, and through new methods including unpublished lists from dozens of experts and through a collaborative effort between botanists and lepidopterists to extract host information from photos of larvae on iNaturalist. When available, we retained all location information.

690

691 We performed the same taxonomic name cleaning and harmonization steps described above for
692 occurrence records, and used all available information on the interaction type to classify
693 interaction types into two categories (“visitsFlowersOf” for pollinator flower visitation records,
694 and “hasHost” for Lepidopteran host plant associations), using reported life stage (i.e., larval or
695 adult) to reclassify ambiguous Lepidopteran interaction records into flower visitation or host
696 plant. Finally, we removed duplicate interaction records and retained only records where both
697 the pollinator and plant genera match with genera found in California, which were the only ones
698 required in subsequent steps. Thus, our final interaction dataset is a binary network of all
699 observed interactions for genera native to California regardless of location.

700

701 Predicting interaction networks

702 We predicted unobserved plant-pollinator interaction networks for each of California’s 35 Jepson
703 districts by integrating phylogenetic information from known interactions, spatial co-occurrence,
704 and phenological overlap of relevant life stages. We separated this workflow into two interaction
705 types – flower visitation and Lepidopteran larval host plant use – because host plant use is
706 independent of flowering phenology. For flower visitation interaction networks, we used multiple
707 means to constrain the matrix of potential interactions to remove highly implausible pairings
708 (forbidden interactions), each successively adding more interactions but with less certainty
709 around the plausibility of those relationships. In the simplest sense, for pollinator species with at
710 least one interaction record, we filtered potential interactions to all species within a plant genus
711 with which that pollinator had previously been observed to interact with. Separately, we inferred
712 potential interactions using graph embedding and transfer learning (described below). Unless
713 otherwise noted, all analyses presented in this paper relied on predicted interactions for
714 potential plant partners identified using the simple phylogenetic constraint for pollinators with
715 some existing interaction data and from graph embedding and transfer learning for pollinators
716 with no previous interaction data. Our predicted interaction networks are available at
717 doi.org/10.5281/zenodo.19338963.

718

719 Graph embedding and transfer learning.

720 For a given taxon group (e.g., bees), we inferred potential interactions between all native
721 pollinator species that have phylogenetic information and plant genera that have at least one
722 previous interaction record with this taxon group. To do so, we designed a workflow which is
723 adapted from Strydom et al.¹³⁰. Our workflow uses the machine learning technique graph
724 embedding to convert an interaction network (i.e., a network of nodes and edges) into a
725 continuous multi-dimensional space, where we can simplify to a lower dimensional
726 representation of the network⁷¹. Every node (every pollinator and plant genus from the
727 interaction data) thus has a vector of numerical values in this reduced space (via t-SVD¹³¹).
728 Using their vectors, we can calculate the dot product between every pair of pollinator species
729 and plant genus. A dot product measures how similar two vectors are; species with similar
730 interaction patterns in the network will have a high value. We refer to this value as the
731 quantitative index of interaction potential.

732

Using the quantitative index, we classified each value into an interaction or non-interaction based on a threshold. A quantitative index above the threshold indicates a potential interaction, and one below indicates a potential non-interaction. To obtain this threshold, we compared observed interactions (those used to construct the network) to the predicted quantitative index, and the threshold is the value of the quantitative index that maximizes the number of observed interactions predicted to occur while also minimizing the unobserved interactions (i.e. maximizing the sum of true positive and true negative rates based on the classification result)¹³². Note that unobserved interactions could include both true non-interactions and interactions simply not detected during sampling, as there was no way to distinguish between the two.

Using this threshold, we can predict new interactions for pollinator species through two approaches depending on whether the pollinators had any observed interactions. First, for pollinators with no observed interaction data that are known to exist in the region, we assumed that more closely related pollinator species share more similar plant visitations and predicted their interactions via transfer learning (out-of-sample learning). In this transfer learning approach, we first subset a phylogeny for all species present in the checklist, based on existing phylogenies. Then, we used ancestral character estimation to reconstruct the quantitative index of interaction potential for the species without interaction data¹³⁰. This resulted in a set of potential interactions between pollinator species and all species within a plant genus. Second, for pollinator species that already had interaction data, we can readily calculate additional quantitative indices of interaction potential in the interaction network that were previously unobserved (in-sample learning). While the results shown throughout only use the interactions predicted from the transfer learning, for completeness, we predicted an additional 406,892 interaction pairs for pollinators that had some previous interaction data available. We present the additional predicted interactions from in-sample learning in Supplementary Material 2, Tables S11 and in the Zenodo data repository.

We implemented this workflow in Julia¹⁰⁵ using `SpeciesInteractionNetworks.jl`¹³³ to conduct the graph embedding, and `PhyloNetworks.jl`¹³⁴ to conduct the ancestral character estimation. To provide the phylogenetic information required for the ancestral character estimation, we subsetting the Open Tree of Life¹³⁵ according to the pollinator species in the checklist and interaction data. While some groups, like bees and butterflies, have better resolved phylogenies (e.g. *Henríquez-Piskulich et al. (2024)*¹³⁶), we decided to use the Open Tree of Life because it allowed us to maintain a consistent workflow for all pollinator groups, and provided a standardized way to graft tips into the phylogeny. Branch lengths of all phylogenies were computed using Grafen's method¹³⁷ with a power transformation of 0.5. We omitted species that lack phylogenetic information from the Open Tree of Life in the transfer learning, which means species that lacked both observed interactions and phylogenetic information were excluded in transfer learning. To classify the quantitative index of interaction potential into interaction and non-interaction from transfer learning, we identified an average threshold by performing a 10-fold cross-validation (90% of pollinator species in training, 10% in validation). This process was only done for species with observed interactions, and we used a rank that captures 100% variance in t-SVD from the observed interactions, meaning no observed interactions are lost in the training data. Within each fold, we identified a threshold that maximizes the number of

observed interactions predicted to occur while also minimizing the unobserved interactions in the validation data. The final threshold was the average threshold value across the 10 cross-validations (Supplementary Material 2 Tables S3-6). To evaluate the performance of using this average threshold in transfer learning, we performed a 3-fold cross-validation (90% of pollinator species in training, 10% in validation) to calculate the recall rate (true positives / (true positives + false negatives)) in the validation data for each fold (Supplementary Material 2 Tables S7-10). A higher recall rate indicates that more true interactions are successfully recovered. Supplementary Material 2 Table S11 summarizes the overall performance of graph embedding and transfer learning in this study.

786

787 **Flower visitation interaction networks.**

788 For all potential flower visitation interactions—determined either by the observed interaction data, 789 the simple taxonomic rule, or the graph embedding—we calculated the plausibility that pollinator i 790 and plant j interact in ecoregion k as

791

$$792 \quad p_{ijk} = p_{co-occ} * p_{co-phen} = \sum_{s \in k} (p_{occ_{is}} * p_{occ_{js}}) / area_k * [|D_i \cap D_j| / |D_j|], \quad (Eq. 1)$$

793

794 where $p_{occ_{is}}$ and $p_{occ_{js}}$ are the plant and pollinator occupancy probabilities for each grid cell site s 795 (derived from the species distribution models produced in Step 4), $area_k$ is the total area the 796 ecoregion k , and D_i and D_j are the set of days that we estimate the pollinator i to be active and 797 plant j to be flowering using the phenometrics produced in Step 5. We calculated the 25th 798 percentile of all p_{ijk} values for previously confirmed interactions (where pollinator i was observed 799 interacting with plant j in ecoregion k) and used this as a threshold to classify potential but 800 previously unobserved interactions ($p_{ijk} > \text{threshold}$).

801

802 **Lepidopteran host plant interaction networks.**

803 We used a more conservative approach to predict Lepidopteran host interactions, due to the 804 high degree of host specificity and the strong phylogenetic conservatism in these interactions. 805 Here, we restricted predictions to spatially explicit interactions only for plants in genera already 806 documented as hosts for a given Lepidopteran species because most Lepidoptera host plant 807 information is available at the genus level. We calculated the plausibility of Lepidoptera species i 808 using plant j as a host in ecoregion k as p_{co-occ} , the normalized spatial co-occurrence term 809 alone, since host plant use can occur across the year and is independent of the flowering 810 phenology term used for flower visitation. As with flower visitation interactions, we applied a 811 threshold based on the 25th percentile of p_{ijk} values for previously confirmed host interactions to 812 distinguish predicted interactions from those unlikely to occur (Supplementary Material 2 Fig. 813 1B).

814

815 **Validating predicted interactions**

We used cross-validation to assess the reliability of our predicted flower visitation interaction networks. We did not validate Lepidopteran host networks using this approach due to the sparsity of spatially explicit host plant interaction data; however, we assume our host plant predictions are reasonable since we restrict them exclusively to known host genera.

820

For flower visitation networks, we iteratively withheld 20% of unique spatially explicit interaction pairs (from the 5,745 spatially explicit interactions that were recorded at the species level for both plant and pollinators, were for species native to California, and had a georeferenced locality in California), predicted spatial interaction networks using Eq. 1, classified interactions as probable or improbable using the thresholding approach described above, and calculated the recall rate ($TP/(TP+FN)$) for the withheld interaction pairs. We compared the recall rate from our predictive methodology to a random selection of interactions from both the full potential interaction pool (any combination of plant and pollinator species co-occurring in that region) and the filtered interaction pool (after removing forbidden links based on previous pollinator species-plant genera interaction pairs). We used recall rate rather than accuracy because we cannot distinguish true negatives (interactions that genuinely do not occur) from interactions that simply have not yet been observed. We excluded the graph embedding step from our validation procedure because withholding a portion of spatially explicit interaction data would be unlikely to eliminate many unique pollinator species–plant genus interaction combinations, instead validating this step independently (see above).

836

837 Comparing to existing planting resources

We compared potential pollinator diversity supported by our data-driven approach against two existing spatially explicit plant recommendation resources—Pollinator Partnership⁷³ (which uses Bailey Ecoregions¹³⁸) and the Xerces Society⁷² (which uses custom boundaries that we digitized in QGIS¹³⁹; see Supplementary Material 2 Fig. S22)—and against random draws from plants native to each region. All comparisons used mixes of 10 plants selected by one of four methods: (1) random native plants, (2) random draws from Pollinator Partnership or Xerces Society regional lists, (3) random draws from NECTAR's top-ranked plants, and (4) a complementary set optimized by NECTAR's genetic algorithm (described below). Because Jepson, Bailey, and Xerces boundaries are not perfectly congruent, we subdivided all regional boundaries into unique combinations of the three classification systems (Jepson eFlora districts, Bailey Ecoregions, and Xerces regions).

849

For methods 3 and 4, we identified top-ranked plants as the N highest-degree species (those supporting the greatest pollinator richness) from predicted flower visitation networks for each Jepson district, where N matched the length of the external plant lists in that region. We then either randomly sampled 10 plants from this list or selected the complementary optimized set using a genetic algorithm. Pollinator support was defined as the number of pollinator species with at least one interaction among the sampled plants.

856

To quantify variance in the estimates of pollinator support across plant selection methods, we fit Bayesian mixed-effects models with brms¹⁴⁰ for each of five pollinator groups (all pollinators, bees, hoverflies, butterflies, and moths) as

860

$$y_s \sim \text{Binomial}(n_s, p_k), \quad (\text{Eq. 2})$$

862

$$\text{logit}(p_s) = \alpha_m + u_k * (1 - d_{\text{external}}) + u_{P2, \text{Xerces}} * d_{\text{external}}, \quad (\text{Eq. 3})$$

864

865 where y_s is the unique number of pollinators in the focal taxonomic group supported by the sth
866 plant sample and n_k is the total number of pollinators for that group in ecoregion k, in which
867 sample s was drawn from. α_m is the fixed intercept for the method m used for selecting that
868 plant mix. Lastly, to control for spatial non-independence of samples drawn, we include u_k ,
869 which is a random effect of Jepson eFlora district k; $u_{P2, \text{Xerces}}$, a multi-membership random effect
870 of the Pollinator Partnership and Xerces Society regions; and d_{external} , a dummy variable
871 indicating which samples were drawn from the external plant lists so the random effects can be
872 appropriately attributed.

873

874 We also ran a separate model where the linear equation was defined as

875

$$\text{logit}(p_s) = \alpha_m + \beta_m \log(x_{\text{density}, k}) + u_k (1 - d_{\text{external}}) + u_{P2, \text{Xerces}} d_{\text{external}}. \quad (\text{Eq. 4})$$

877

878 Eq. 4 includes β_m , a fixed slope of interaction data density in ecoregion k ($x_{\text{density}, k}$) with respect
879 to the method used to select the plant mix in sample s. We included this model separately to
880 assess whether the quality of plant choices varied in any sampling scheme by the initial density
881 of interaction data.

882

883 It is worth noting that using the predicted interaction networks from NECTAR as the basis for
884 comparing plant mixes from different methods assumes plants only support pollinators in
885 regions they are found natively. We also repeated this comparison using a metaweb of observed
886 plant-pollinator interactions (See Supplementary Material 2 Fig. S11), assuming any pollinator
887 recorded in a Jepson district can interact with a sampled plant if that interaction is documented
888 anywhere in collated interaction records. In both instances, we matched external plant lists to
889 the interaction datasets at the species level, meaning we may have overemphasized the value
890 of cultivars suggested by external lists to pollinators, since cultivars often support fewer
891 pollinators than the species they are derived from¹⁴¹.

892

893 Restoration scenarios

894 To test the utility of our approach under realistic conditions, we developed three restoration
895 scenarios, each with distinct objectives and constraints, and compared the pollinator support
896 achieved using our data-driven approach versus random plant selection without interaction data
897 (as our null model) from the ecoregional species pool that otherwise meets the scenario-specific
898 constraints. In each scenario, we identified the top 10 plants that together maximize the stated
899 objective using a genetic algorithm. The genetic algorithm is similar to that proposed by

M'Gonigle et al.⁵¹ and iteratively evolved 200 “populations” of 10 unique plants (among the constrained suitable plant pool) over 200 generations. In each generation, the algorithm created new plant combinations by selecting the high-performing combinations of parents (plant sets from the previous generation), combining the plants from these parent lists, and occasionally replacing plants with randomly selected alternatives (according to a mutation rate of 0.1). We compared these optimized plant lists to the average pollinator support of 100 draws of 10 randomly selected plants from the corresponding Jepson district's species pool that met all planting constraints. For all scenarios, we visualized the proportion or number of species of the focal pollinator community supported by the optimized plant lists compared to random selection in two separate regions: Los Angeles (South Coast district) and Redding (Cascade Range Foothills district). However, we ran the analysis for all regions in California. The list of identified plants and corresponding pollinator support is available in Supplementary Material 3 Tables S1-2.

913

Scenario 1: Chaparral community restoration. This scenario aimed to maximize pollinator diversity using plants suitable for chaparral community restoration that are relatively common and feasible to collect for seed-based restoration. We constrained candidate plants to those associated with chaparral community (using Calflora's plant community association data¹⁰⁷) and excluded rare or endangered species by removing all plants with California Rare Plant Rankings of 1–3⁷⁴. The genetic algorithm's fitness function maximized the number of unique pollinator species (bees, butterflies, moths, and hoverflies) supported by flower visitation interactions across the selected plant combination.

922

Scenario 2: Drought-tolerant specialist bee garden. This scenario optimized support for oligolectic (pollen specialist) bee diversity in a drought-tolerant garden context. We constrained candidate plants to species requiring low water usage and available in at least one California nursery using Calscape plant data¹⁴². We ran two separate optimizations: one that maximized overall pollinator diversity (all flower-visiting species), and another that specifically maximized oligolectic bee diversity. For the oligolectic bee optimization, we used data from Fowler (2020)¹²⁸ to identify which of our predicted interactions were between oligolectic bees and plants in their known host genera.

931

Scenario 3: Restoration to support declining butterflies. This scenario identified plants that best support declining southwestern butterfly species through both larval host plant and adult nectar support. We collated all butterflies classified as "Declining" in the Southwest region from ref.⁷⁵ and used the genetic algorithm to identify plant sets maximizing support for this subset of butterfly species, with the constraint that host and nectar support must come from different plant species (i.e., a butterfly was only considered supported if the plant combination included at least one host plant and at least one distinct nectar plant for that species). Among plant combinations supporting equal numbers of declining butterflies, we selected the set that additionally supported the greatest overall pollinator diversity across all interaction types.

941

Quantifying spatial biases and exploring network structure

We summarised the proportion of observed interactions present in the collated interaction data belonging to different genera of plants and pollinators and compared this to our predicted interaction networks. We visualized the degree of each genus in the raw and predicted interaction networks for California native species by aggregating all unique plant-pollinator pairs that occur anywhere (for the raw data) or in any region (for predicted interaction networks), summarising the number of unique pollinator species to plant species interactions between all combinations of plant and pollinator genera, and then plotting them on a graph with the graph visualization software Gephi⁷⁶.

951

Further, we showed how species and interaction turnover contribute to differences in network structure using the predicted interaction networks for each Jepson district in California. We created maps of expected plant and pollinator diversity across space by summing the relative habitat suitabilities from the ensemble species distribution models (Supplementary Material 2 Figs. S15-16). Further, we calculated and plotted nestedness and mean degree for pollinators for the downscaled regional flower visitation interaction network for each Jepson district (Supplementary Material 2 Figs. S17-18). Lastly, we calculated mean interaction network dissimilarity for each pairwise combination of downscaled regional interaction networks and partitioned the differences into species turnover and interaction rewiring components (Supplementary Material 2 Figs. S19-21)^{143,144}.

962

The code for NECTAR and all analyses presented in this manuscript can be found at github.com/EntoDataSciCornell/NECTAR, alongside the major data products produced at doi.org/10.5281/zenodo.19338963. We used R (v4.2.3)¹⁴⁵ for most analyses. For data handling and manipulation we used tidyverse (v2.0.0)¹⁴⁶, and data.table¹⁴⁷ (v1.16.2). For spatial data handling we used sf (v1.0.21)¹⁴⁸, terra (v1.8.70)¹⁴⁹, raster (v3.6.32)¹⁵⁰, tigris (v2.2.1)¹¹³, maps (v3.4.3)¹⁵¹, rmapshaper (v0.5.0)¹⁵², and ngeo (v0.4.8)¹⁵³. For occurrence data download and access we used rgbif (v3.8.3)¹⁵⁴, ridigbio (v0.4.1)¹⁵⁵, rvest (v1.0.4)¹⁵⁶, and httr (v1.4.7)¹⁵⁷. For occurrence data cleaning we used CoordinateCleaner (v3.0.1)¹⁵⁸, bdc (v1.1.5)¹¹², and BeeBDC (v1.3.1)⁹⁶. For species distribution modelling we used flexsdm (v1.3.9)¹⁵⁹, biomod2 (v4.2.6.2)¹⁶⁰, ecospat (v4.1.2)¹⁶¹, blockCV (v3.2.0)¹⁶², CovSel (v0.1.0)¹⁶³, dismo (v1.3.16)¹⁶⁴, adehabitatHR (v0.4.22)¹⁶⁵, mda (v0.5.5)¹⁶⁶, earth (v5.3.4)¹⁶⁷, geodata (v0.6.6)¹⁶⁸, rnaturalearth (v1.1.0)¹⁶⁹, nhplusTools (v1.3.2)¹⁷⁰, and FedData (v4.3.0)¹⁷¹. For phenology modelling we used circular (v0.5.2)¹⁷² and boot (v1.3.28.1)¹⁷³. For interaction data scraping and preparation we used rvest (v1.0.4)¹⁵⁶ and rotl (v3.1.0)¹⁷⁴. For phylogenetic data handling we used ape (v5.8.1)¹⁷⁵. For interaction prediction and network analysis we used bipartite (v2.23.0)¹⁷⁶ and ngeo (v0.4.8)¹⁵³. For the genetic algorithm plant selection analyses we used futureverse (v1.34.0)¹⁷⁷. For Bayesian statistical modelling and analysis we used brms (v2.23.0)¹⁴⁰, cmdstanr (v0.9.0)¹⁷⁸, tidybayes(v3.0.7)¹⁷⁹. For visualization we also used patchwork (v1.3.0)¹⁸⁰, cowplot (v1.1.3)¹⁸¹, ggdist (v3.3.3)¹⁸², ggalluvial (v0.12.5)¹⁸³, tidyterra (v0.7.2)¹⁸⁴, and units (v0.8.5)¹⁸⁵. For reading external data files we used readxl (v1.4.3)¹⁸⁶. For the graph embedding and transfer learning, we implemented them in Julia (v1.10.10)¹⁰⁵ using packages CSV.jl (v0.10.15)¹⁸⁷, DataFrames.jl (v1.7.0)¹⁸⁸, Distributions.jl (v0.25.120)¹⁸⁹, JLD2.jl (v0.6.2)¹⁹⁰, PhyloNetworks.jl (v0.16.4)¹³⁴, Pipe.jl (v1.3.0)¹⁹¹, Random.jl (a core standard library in Julia)¹⁰⁵, SpeciesInteractionNetworks.jl (v0.0.1)¹³³, and StatsBase.jl (v0.34.6)¹⁹².

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989 References

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1403

1404 **Supplementary**

1405 *Supplementary Material 1* - Checklist of native plants and pollinators that could potentially have

1406 been included in NECTAR, including the 5,178 plant and 5,131 pollinator species that we

1407 include in NECTAR's interaction prediction (species which we were able to run SDMs and

1408 phenometric estimations for).

1409 *Supplementary Material 2* - Additional figures and tables

1410 *Supplementary Material 3* - Optimized plant lists for each restoration scenario presented and the

1411 corresponding pollinator support scores.

1412

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